

1 Prof's Blueprint (Final)

Exam: MCQ + written, **30 pts total**. Calculator allowed (phones/tablets no); **no programming** required. Practical problems dominate (quiz/HW-like); fewer theory. One A4 sheet (this one).

Traps: Euler needs $\gcd(a, n)=1$ (FLT needs p prime too); power cipher needs $bd \equiv 1 \pmod{p-1}$; product $\deg n+m$ needs $n+m+1$ points; $N_{\text{FFT}} =$ smallest $\text{pow}2 \geq 2n-1$; GCD is monic (lead 1); $\deg h=0 \Rightarrow$ inconsistent system.

2 Groups Recap + Euler φ

Group: closure, identity e (unique, $aee=a$), associativity, inverses (unique). $(\mathbb{Z}, +)$, $(\mathbb{Z}_n, +)$ groups; $\mathbb{Z}_n^*$ group $\iff n$ prime. $|G|=\#\text{elem}$; $\text{ord}(g) =$ smallest $n>0$ with $g^n=e$. **Lagrange:** $\text{ord}(g) \mid |G|$, $g^{|G|}=e$. **Worked:** $|\mathbb{Z}_8^+|=8$, $\text{ord}[1]=8$, $\text{ord}[2]=4$, $\text{ord}[7]=8$; gens $[1], [3], [5], [7]$. $\varphi(n)=\#\{1 \leq k \leq n : \gcd(k, n)=1\}$. **Formulas:** p prime $\varphi(p)=p-1$; $\varphi(p^k)=p^k - p^{k-1}$; coprime $\varphi(mn)=\varphi(m)\varphi(n)$. **Worked:** $\varphi(16)=8$ ($1, 3, 5, 7, 9, 11, 13, 15$); $\varphi(17)=16$; $\varphi(14)=\varphi(2)\varphi(7)=(2-1)(7-1)=1 \cdot 6=6$; $\varphi(9)=9-3=6$ (prime power).

Euler: $a^{\varphi(n)} \equiv 1 \pmod{n}$ if $\gcd(a, n)=1$. FLT = prime case ($\varphi(p)=p-1$) = Lagrange on $\mathbb{Z}_p^*$. **Difference:** Euler works for composite moduli too.

Worked $3^{347} \bmod 14$: Given $\gcd(3, 14)=1$, $\varphi(14)=6$.

Step 1 (reduce exponent): $347=6 \cdot 57+5$ ($6 \cdot 57=342$).

Step 2 (evaluate): $3^5=243$;

$243-17 \cdot 14=243-238=5 \Rightarrow$ [5]. **Check:** $\text{pow}=5$.

3 Crypto

Caesar: $c \equiv m+s \pmod{26}$, $m \equiv c-s$. **Worked** BY ($B=1, Y=24$), $s=20$: $1+20=21 \equiv V$, $24+20=44-26=18 \equiv S \Rightarrow$ **VS**.

Power: $c \equiv m^b \pmod{p}$; $m \equiv c^d$, $bd \equiv 1 \pmod{p-1}$. **Worked** $p=29$, $b=5$: $B: 1^5=1 \equiv B$.

Y Step 1 ($24 \equiv -5$): $(-5)^2=25$; **Step 2:** $25^2=625$; $625-21 \cdot 29=625-609=16$.

Y Step 3: $16 \cdot (-5)=-80$;

$-80+3 \cdot 29=-80+87=7 \Rightarrow$ **H**. $d: 5d \equiv 1$

$\pmod{28} \Rightarrow$ **d=17** ($5 \cdot 17=85=3 \cdot 28+1$).

Correctness = bijective: $D(E(x))=x$, $E(D(y))=y$. Caesar always bijective (shift). Power bijective $\iff \gcd(b, p-1)=1$ (else many-to-one). Many-to-one: ≥ 2 symbols map together; surjective: covers whole target.

Check decrypt: $VS-20$: $21-20=1 \equiv B$, $18-20=-2+26=24 \equiv Y$. Power: $7^{17} \bmod 29=24 \equiv Y$ **Check OK**.

4 Complex Numbers

Rect $z=a+bi$ ($\text{Re}=a$, $\text{Im}=b$), plane vector (a, b) . Polar $r(\cos \theta + i \sin \theta)$, $r=|z|=\sqrt{a^2+b^2}$. $|z|^2=z\bar{z}$, $\bar{z}=a-bi$.

$+$, $-$ = vector + (parallelogram). Mult: $(a+bi)(c+di)=(ac-bd)+(ad+bc)i$. Div: $\times^{-1}$: $z_1/z_2=z_1\bar{z}_2/|z_2|^2$. Inverse $z^{-1}=\bar{z}/|z|^2$.

Worked $3+4i$: $|z|=5$,

$z^{-1}=(3-4i)/25=0.12-0.16i$.

Worked $(1+6i)/(1-6i)$:

$(1+6i)^2/37=(-35+12i)/37$.

$\approx -0.9459+0.3243i$ **Check** ($37 \cdot (-0.9459)=-35$, $37 \cdot 0.3243=12$).

Euler: $e^{i\theta} = \cos \theta + i \sin \theta$; **DeMoivre:** $(re^{i\theta})^n = r^n(\cos n\theta + i \sin n\theta)$.

n -th roots: $\sqrt[n]{r}(\cos(\theta+2k\pi)/n + i \sin(\theta+2k\pi)/n)$, $k=0..n-1$.

Polar mult: $r_1 r_2 e^{i(\theta_1+\theta_2)}$ (lengths $\times$, angles $+$, mod 2π): $e^{i\pi/3} \cdot e^{i\pi/6} = e^{i\pi/2} = i$.

Worked $8e^{i\pi/3}$: $8(\frac{1}{2}+i\frac{\sqrt{3}}{2})=4+4\sqrt{3}i \approx 4+6.9282i$.

Worked $11e^{3\pi i/2}$: $11(0-i)=-11i$.

Cube roots of $e^{i\pi/2}$: angles

$\pi/6, 5\pi/6, 3\pi/2 \Rightarrow \sqrt{3}/2+i/2, -\sqrt{3}/2+i/2, -i$.

Square roots of $e^{i\pi/4}$: angles $\pi/8, 9\pi/8$:

$\cos \frac{\pi}{8} = \sqrt{2+\sqrt{2}}/2 \approx 0.9239$,

$\sin \frac{\pi}{8} = \sqrt{2-\sqrt{2}}/2 \approx 0.3827$.

Rect $\rightarrow$ polar: $2+2i$: $r=\sqrt{8}=2\sqrt{2}$,

$\theta=\pi/4 \Rightarrow 2\sqrt{2}e^{i\pi/4}$ (**Check**

$2\sqrt{2}(\sqrt{2}/2+i\sqrt{2}/2)=2+2i$).

Geometry: $\times e^{i\theta}$ rotates by θ (lengths mult); unit circle $\{|z|=1\}$ subgroup (closed $|zw|=1$, $\text{inv } \bar{z}$).

Plot recipe: mark Re on x , Im on $y \rightarrow$ arrow; add tip-to-tail; mult adds angles, mults lengths.

Groups: $(\mathbb{C}, +)$ $e=0$ $\text{inv } -z$; $(\mathbb{C}^*, \cdot)$ $e=1$ $\text{inv } \bar{z}/|z|^2$;

$C_4^*=\{\pm 1, \pm i\}$ gen i ; C_8^* gen $(1+i)/\sqrt{2}$; $\{1, -1\}$

group (gen -1); $\{i, -i\}$ **not** ($i \cdot i=-1$ missing).

Mult prop $|z_1 z_2|=|z_1||z_2|$: $z_1=(1+i)/\sqrt{2}$,

$z_2=(1-i)/\sqrt{2}$: $1 \cdot 1=|1|=1$ **Check OK**.

5 Roots of Unity

$\omega_n = e^{2\pi i/n}$; powers $\omega_n^0.. \omega_n^{n-1} =$ all n roots

cyclically; primitive = generator; n distinct roots of $z^n-1=0$.

Primitive: ω_n^k generates all $\iff \gcd(k, n)=1$;

e.g. $n=8$: $k=1, 3, 5, 7$.

Relations (n even): $\omega_n^n=1$, $\omega_n^{n/2}=-1$, $\omega_n^2=\omega_{n/2}$, $\omega_n^{-1}=\bar{\omega}_n$, $\omega_n^{k+n/2}=-\omega_n^k$, $\omega_n^{2k}=\omega_{n/2}^k$, $\omega_n^{-k}=\bar{\omega}_n^k$.

Worked $\omega_8=(1+i)/\sqrt{2}$: $\omega_8^4=-1$, $\omega_8^8=1$,

$\omega_8^{-5}=\omega_8^{-5+8}=\omega_8^3=(-1+i)/\sqrt{2}$;

$\omega_{16}^8=e^{8\pi i/16}=i=\omega_4$. $\sqrt{i}=(1+i)/\sqrt{2}$ (check

$((1+i)/\sqrt{2})^2=i$); $\sqrt{-i}=(-1+i)/\sqrt{2}$.

Eval trick: reduce exponents mod n , use $\omega_2=-1$.

Worked $p=x^4-3x^3+x^2-2x-1$ at $\omega_4=i$: powers

$\omega_4^4=1$, $\omega_4^3=-i$, $\omega_4^2=-1$.

Then

$1-3(-i)+(-1)-2i-1=1+3i-1-2i-1=-1+i$.

Lecture: $3\omega_4^3-2\omega_4^2+2+\omega_4^8=3(-i)+2+2+1=5-3i$.

6 Iso $C_n^* \cong \mathbb{Z}_n^+$

Map $\omega_n^k \leftrightarrow [k]_n$; $\times \leftrightarrow +$:

$\omega_n^k \omega_n^l = \omega_n^{k+l} \leftrightarrow [k]+[l]=[k+l]$. Gen $\leftrightarrow$ gen.

Worked $C_4 \leftrightarrow \mathbb{Z}_4$: $[0] \leftrightarrow 1$, $[1] \leftrightarrow i$, $[2] \leftrightarrow -1$, $[3] \leftrightarrow -i$;

$[1]+[2]=[3] \leftrightarrow i \cdot (-1)=-i$;

$[3]+[2]=[5]=[1] \leftrightarrow (-i)(-1)=i$. Tables same shape

$\Rightarrow$ iso.

No-iso test: $|G|$ must match + generator orders;

e.g. $|C_2^*|=2$ vs $|\mathbb{Z}_4^+|=4 \Rightarrow$ no iso. **Fill C_5^* via iso:**

$\omega_5^3 \omega_5^4 = \omega_5^7 = \omega_5^2 \leftrightarrow [3]+[4]=[7]=[2]$. Subgroups:

$C_{2^m}^* \subset C_{2^{m+1}}^*$ ($z^{2^m}=1 \Rightarrow (z^{2^m})^2=1$).

7 Poly: FTA, Fields, Coeff/Value

Field: $+$, $-$, $\times$, $/$ (no $/0$); $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ yes; $\mathbb{Z}, \mathbb{N}$ no. Alg closed: every $\deg \geq 1$ has a root; $\mathbb{C}$ is (FTA).

FTA: $\deg n \Rightarrow$ **exactly n roots in $\mathbb{C}$** counting mult: $f=a_n \prod (x-x_i)$.

Worked $x^4+16=0$: no real roots; 4 in $\mathbb{C}$; two:

$\sqrt{2}+i\sqrt{2}$, $-\sqrt{2}+i\sqrt{2}$ (check $(\sqrt{2}+i\sqrt{2})^2=4i$,

$^2=-16$). Factor thm: a root $\iff (x-a) \mid f$;

$(x-1)^2=x^2-2x+1$ (mult 2). Note: 5 prime in $\mathbb{Z}$

but $5=(2-i)(2+i)$ in $\mathbb{Z}[i]$.

Coeff $(a_0..a_n)$; **value:** $\deg \leq n$ unique from $n+1$

pts (Vandermonde $\det \neq 0$). **Product $\deg n+m$**

needs $n+m+1$ pts; pad zeros

$(4+5x \rightarrow 4+5x+0x^2)$. Pointwise

$h(x)=f(x)g(x)$, $O(m)$; schoolbook $O(n^2)$.

Worked $f=1-x$, $g=2+x$ at $-2, 0, 1$:

$f: 1-(-2)=3, 1, 0$; $g: 2+(-2)=0, 2, 3$;

$h: 0, 2, 0 \Rightarrow h=2-x-x^2$ (check $2+2-4=0$, 2 ,

$2-1-1=0$).

Sketch recipe: mark roots $\rightarrow$ lead-coeff sign for ends $\rightarrow y$ -intercept $\rightarrow$ join; graphical mult: sample ≥ 6 x 's, multiply y 's, plot.

8 FFT/IFFT + Costs

Even/odd: $p(x)=p_e(x^2)+x p_o(x^2)$. **Worked**

$1-2x+3x^2-4x^3+5x^4-6x^5+7x^6-8x^7$:

$p_e(y)=1+3y+5y^2+7y^3$, $p_o(y)=-2-4y-6y^2-8y^3$;

depth of recursion tree $=3$ ($8=2^3$).

Butterfly ($k=0..n/2-1$):
 $p(\omega^k)=p_e(\omega^{2k})+\omega^k p_o(\omega^{2k})$;
 $p(\omega^{k+n/2})=p_e(\omega^{2k})-\omega^k p_o(\omega^{2k})$. Needs $n=2^t$.

$N_{\text{FFT}} \geq 2n-1$, smallest $\text{pow}2$ ($4+4$ coeffs $\deg 3$:

product $\deg 6$, 7 coeffs $\Rightarrow$ **$N=8$**). Pipeline:

coeff $\rightarrow$ FFT $\rightarrow$ vals $\rightarrow$ pointwise $\times \rightarrow$ IFFT $\rightarrow$ coeff.

Zero-pad; extra points OK if same poly. **Idea:** one

(p_e, p_o) pair gives two values (halves work).

IFFT: $a_j = \frac{1}{n} \sum_{k=0}^{n-1} y_k (\omega^{-j})^k$. **Costs:**

FFT/IFFT $O(n \log n)$ (depth $\log n \times O(n)/\text{level}$);

pointwise $O(n)$; total $O(n \log n)$.

Worked FFT $q=1-x^2+4x^3$ ($N=4$, $\omega_4=i$):

$q(1)=4$; $q(i)=2-4i$; $q(-1)=-4$; $q(-i)=2+4i$.

Butterfly check: $p_e(y)=1-y$, $p_o(y)=4y$;

$q(i)=p_e(-1)+i p_o(-1)=2+i(-4)=2-4i$.

Butterfly table: $k=0$: $p_e(1)=0$,

$p_o(1)=4 \Rightarrow q(1)=0+4=4$, $q(-1)=0-4=-4$.

$k=1$: $p_e(-1)=2$,

$p_o(-1)=-4 \Rightarrow q(i)=2+i(-4)=2-4i$,

$q(-i)=2-i(-4)=2+4i$.

Full pipeline $f=1-x$, $g=2+x$ ($N=4$):

$F: 0, 1-i, 2, 1+i$; $G: 3, 2+i, 1, 2-i$;

$H=F \cdot G: 0, 3-i, 2, 3+i$.

IFFT: $a_0=(0+3-i+2+3+i)/4=8/4=2$;

$a_1=(-3i-1-2+3i-1)/4=-1$;

$a_2=-1 \Rightarrow h=2-x-x^2$ **Check OK**.

Worked IFFT $y=(-1, i, 1, 2)$:

$a_0=(-1+i+1+2)/4=(2+i)/4=0.5+0.25i$;

$a_1=(-1+1-1+2i)/4=(-1+2i)/4=-0.25+0.5i$.

$a_2=(-1-i+1-2)/4=(-2-i)/4=-0.5-0.25i$;

$a_3=(-1-1-1-2i)/4=(-3-2i)/4=-0.75-0.5i$.

Check $p(i)=a_0+a_1 i-a_2-a_3 i=y_1$.

9 Int Mult, GCD, Systems

Int as poly base B : $325 \rightarrow 5+2x+3x^2$ ($p(10)=325$).

Steps: polys $\rightarrow$ FFT $\rightarrow$ pointwise $\times \rightarrow$ IFFT $\rightarrow$

carry to $[0, B-1] \rightarrow$ eval B . **Carry:**

$[2, 15, 7] \rightarrow [2, 5, 8]=852$ (raw $2+150+700=852$;

notes typo 825). $5678 \rightarrow 8+7x+6x^2+5x^3$;

$q(10)=5678$, $q(2)=86$.

GCD monic (lead 1): $4x^2+12x-7 \rightarrow x^2+3x-7/4$.

Euclid via synthetic division.

Worked $f=x^4-8x^3+17x^2+2x-24$,

$g=x^3-6x^2+11x-6$:

Step 1 (divide): $f-(x-2)g=-6x^2+30x-36=-6h$,

$h=x^2-5x+6$.

Step 2: $g \bmod h$ is 0 $\Rightarrow \gcd(f, g)=h=x^2-5x+6$.

Synthetic $(x^3-6x^2+11x-6)/(x-1)$: bring 1; $\times 1$:

$1 \rightarrow -5$; $\times 1$: $-5 \rightarrow 6$; $\times 1$: $6 \rightarrow 0$.

Quot x^2-5x+6 , rem 0 (so $x=1$ root).

Overdetermined: $h=\gcd$; $\deg h=0 \Rightarrow$

inconsistent ($x-1=0$, $x+1=0$); $\deg \geq 1 \Rightarrow$ common roots = roots of h .

Worked: $x^2-11x+30=0$: $25-55+30=0$,

$36-66+30=0 \Rightarrow 5, 6$.

$x^3-10x^2+27x-18=0$: test 6:

$216-360+162-18=0$; roots 1, 3, 6 $\Rightarrow$ common $x=6$.

Resultant: det condition for common root

(eliminates a variable). 2×2 :

$\text{Res}(x-1, x+1)=(1)(1)-(-1)(1)=2 \neq 0 \Rightarrow$ no

common root.

10 Extra Worked Q&A (Quiz/HW Style)

FFT size Qs: p with n coeffs needs n points

Inverse of $(-1-i)/\sqrt{2}$: $|z|^2=1$, $\text{inv} = \bar{z}=(-1+i)/\sqrt{2}=z_6$.

Group tests: $2a+7bi$ ($a, b \in \mathbb{Z}$) additive: closed, $e=0$, $\text{inv } -z \Rightarrow$ **group**. $3a+3bi$, $ab \neq 0$ multiplicative: $1=3a+3bi \Rightarrow a=1/3 \notin \mathbb{Z} \Rightarrow$ **not** (no identity).

FFT $n=32$: depth 5 ($2^5=32$); butterfly pairs (ω^0, ω^{16}) , (ω^1, ω^{17}) ; even/odd parts deg 15, split again $\rightarrow$ 4 polys $(p_{ee}, p_{eo}, p_{oe}, p_{oo})$. C_n^+ = cyclic additive order n ($\cong \mathbb{Z}_n^+$), gens $[k]$ with $\gcd(k, n)=1$; chain $C_{2^m}^* \subset C_{2^{m+1}}^*$.

Inconsistent check: $\gcd(x-1, x+1)=1$ (deg 0) $\Rightarrow$ **inconsistent** $((x+1)-(x-1)=2 \neq 0)$.

11 —Worked Drills: Mod + Groups/FLT

Reduce first: $[125]_{17}+[241]_{17}=[366]_{17}$; $366-21 \cdot 17=366-357=9$. $[391]_{21} \cdot [457]_{21}$: $391-18 \cdot 21=13$, $457-21 \cdot 21=16$; $13 \cdot 16=208-9 \cdot 21=19$. $[9^4]_7$: $9 \equiv 2$, $2^4=16 \equiv 2$. $([157]_{13}+[249]_{13}) \cdot [83]_{13}$: $157-12 \cdot 13=1$, $249-18 \cdot 13=249-234=15$; $15-13=2$; $1+2=3$; $83-6 \cdot 13=5$; $3 \cdot 5=15-13=2$.

HW08: $[123]_{17}+[245]_{17}=11$ ($368-357$). $[391]_{23} \cdot [457]_{23}=0$ ($391=17 \cdot 23$). $([156]_{11}+[247]_{11}) \cdot [83]_{11}$: $2+5=7$, $\times 6=42-33=9$. $[312]_{15} \cdot ([184]_{15}+[257]_{15})$: $12 \cdot (4+2)=72-60=12$.

Find x : $143 \equiv x \pmod{13}$: $143=11 \cdot 13 \Rightarrow 0$. $12345 \bmod 11$: $11 \cdot 1122=12342 \Rightarrow 3$. $54321 \bmod 7$: $7 \cdot 7760=54320 \Rightarrow 1$. $147 \bmod 13$: $13 \cdot 11=143 \Rightarrow 4$.

Find n : $24 \equiv 0 \pmod{n} \Rightarrow n|24$: $1, 2, 3, 4, 6, 8, 12, 24$. $15 \equiv 1 \Rightarrow n|14$: $1, 2, 7, 14$. $124 \equiv 4 \Rightarrow n|120$, $n>4$: $5, 6, 8, 10, 12, 15, 20, 24, 30, 40, 60, 120$. $18 \equiv 0$, $0 < n \leq 18$: $1, 2, 3, 6, 9, 18$.

Classes: mod 7: 7 classes $[0] \dots [6]$. $[2]_5=\{\dots, -8, -3, 2, 7, 12, \dots\}$; $[4]_7=\{\dots, -3, 4, 11, \dots\}$; $[-3]_8=[5]_8=\{\dots, -3, 5, 13, \dots\}$. $\{1, -1\} \subset C_4^*$ subgroup (closed, $e=1$, self-inv).

Groups: $(\mathbb{Q}, +)$ group; $(\mathbb{Q}, \cdot)$ **not** (0 no inverse). $(\mathbb{Z}_4, +)$ group ($e=[0]$, $\text{inv } [-a]$). $\mathbb{Z}_4^*=\{1, 2, 3\}$ **not** ($2 \cdot 2=0$). Mult groups on $\mathbb{Z}_n$: $n=3, 5, 13, 17$ yes ($|G|=2, 4, 12, 16$); $10, 15$ no. Possible orders:

$2 \rightarrow \{1, 2\}$; $4 \rightarrow \{1, 2, 4\}$; $12 \rightarrow \{1, 2, 3, 4, 6, 12\}$; $16 \rightarrow \{1, 2, 4, 8, 16\}$. $|\mathbb{Z}_{24}^*|=\varphi(24)=8$ (odd, not mult of 3: $1, 5, 7, 11, 13, 17, 19, 23$); x odd (e.g. $3, 5, 7$) never gives $[a]^x=[1]$ for $a=2 \dots 23$ (needs $2|x$; $[5]^2=[1]$ so every even x works).

FLT big powers: $19^6 \bmod 7$ OK ($7 \nmid 19$); 21^6 **no** ($\gcd=7$); $11^8 \bmod 9$ **no** (9 composite). 19^{68} : $19 \equiv 5$, $68=6 \cdot 11+2$ ($66+2$), $5^2=25 \equiv 4$. $271^{24} \bmod 11$: $271-24 \cdot 11=271-264=7$, $24=10 \cdot 2+4$, $7^4=2401-218 \cdot 11=2401-2398=3$. $290^{293} \bmod 23$: $290-12 \cdot 23=290-276=14$, $293=22 \cdot 13+7$ ($286+7$), $14^7 \equiv 19$. No-FLT: $[16^4]_7=[2]$; $[10^{100}]_9=[1]$; $[23^2]_9=[7]$ ($23 \equiv 5$, $25-18$); $[42]_{12}=[6]$.

12 —Worked Drills: Complex + FFT Sums (plot tip-to-tail, check):

$(-2+4i)+(-3-7i)=-5-3i$; $(-2-i)+(2+i)=0$; $(1-2i)+(1+2i)=2$. $|z|^2=z\bar{z}$: $(-2-3i)$: $4+9=13$; $4i$: 16 ; $\bar{z}=4i \Rightarrow z=-4i$, $|z|^2=16$.

Div: $(2+6i)/2=1+3i$. $(3+4i)/(1-6i)=(3+4i)(1+6i)/37=(-21+22i)/37$. $\approx -0.5676+0.5946i$ **Check** $37 \cdot (-0.5676)=-21$, $37 \cdot 0.5946=22$. **vs dot:** $z_1 z_2=(ac-bd)+(ad+bc)i$ vs $u \cdot v=ac+bd$; $|z_1 z_2|^2=(a^2+b^2)(c^2+d^2)$ vs $(u \cdot v)^2=(ac+bd)^2$; coincide $\iff ad=bc$. E.g. $z_1=1+2i, z_2=3+4i$: $|z_1 z_2|^2=(1+4)(9+16)=125$ vs $(3+8)^2=121$ ($1 \cdot 4 \neq 2 \cdot 3$).

Even/odd twice: $p_e(y)=1+3y+5y^2+7y^3 \Rightarrow p_{ee}(z)=1+5z$, $p_{eo}(z)=3+7z$; $p_o(y)=-2-4y-6y^2-8y^3 \Rightarrow p_{oe}(z)=-2-6z$, $p_{oo}(z)=-4-8z$. **Check:** $p_{ee}(y^2)+y p_{eo}(y^2)=1+5y^2+3y+7y^3=p_e$.

Big-O tree (n leaves, balanced binary): depth $\log_2 n$; bottom-only work $\Rightarrow O(n)$; $O(n)$ per level $\Rightarrow O(n \log n)$.

C_5^* row via iso: $\omega_5^4 \cdot \omega_5^4=\omega_5^8=\omega_5^3 \leftrightarrow [4]+[4]=[8]=[3]$. Gen $\omega_5 \leftrightarrow [1]_5$ powers fill table.

13 —Worked Drills: Tables + FFT

Mod-8 $\times$ table (nonzero rows): Rule: row a has 1 $\iff a$ invertible; row a hits 0 $\Rightarrow$ zero divisor. Units = odd $\{1, 3, 5, 7\}$ ($\gcd=1$); $1^{-1}=1$, $3^{-1}=3$ ($9-8=1$), $5^{-1}=5$ ($25-24=1$), $7^{-1}=7$ ($49-48=1$). Even rows lack 1. **Zero divisors:** $2 \cdot 4=8 \equiv 0$, $6 \cdot 4=24 \equiv 0$ ($24-24$).

FFT $f=2+3x+5x^3$ ($N=4$, $\omega_4=i$; $p_e(y)=2$, $p_o(y)=3+5y$): $f(1)=2+3+5=10$; $f(i)=2+3i+5(-i)=2-2i$; $f(-1)=2-3-5=-6$; $f(-i)=2-3i+5i=2+2i$.

DeMoivre power: $((1+i)/\sqrt{2})^8=(e^{i\pi/4})^8=e^{2\pi i}=1$ (angle $\times 8$). Specials: $e^{i\pi}=-1$, $e^{2\pi i}=1$. **Decrypt proof:** $m \equiv c^d=(m^b)^d=m^{bd}=m^{1+k(p-1)}=m \cdot (m^{p-1})^k \equiv m$ (FLT, p prime, $p \nmid m$).

14 —Tables + Step-by-Step Recoveries

Mod-8 $\times$ table (read inverses/zero divisors; odd rows permute, even rows hit 0):

$\times$	1	2	3	4	5	6	7
1	1	2	3	4	5	6	7
2	2	4	6	0	2	4	6
3	3	6	1	4	7	2	5
4	4	0	4	0	4	0	4
5	5	2	7	4	1	6	3
6	6	4	2	0	6	4	2
7	7	6	5	4	3	2	1

C_4^* vs $\mathbb{Z}_4^+$ (same shape $\Rightarrow$ iso $\omega_4^k \leftrightarrow [k]_4$):

*	1	i	-1	$-i$	+	0	1	2	3
1	1	i	-1	$-i$	0	0	1	2	3
i	i	-1	$-i$	1	1	1	2	3	0
-1	-1	$-i$	1	i	2	2	3	0	1
$-i$	$-i$	1	i	-1	3	3	0	1	2

Recover $h=a+bx+cx^2$ from values $(0, 2, 0)$ at $(-2, 0, 1)$: $a-2b+4c=0$, $a=2$, $a+b+c=0$. **Step 1:** $a=2$. **Step 2:** $b+c=-2$, $-2b+4c=-2 \Rightarrow b-2c=1$. **Step 3:** subtract: $3c=-3 \Rightarrow c=-1$, $b=-1$. **Box:** $h=2-x-x^2$. **Check:** $2+2-4=0$, $2, 2-1-1=0$. **Read graph $\rightarrow$ number:** point $(x, y) \rightarrow x+yi$; length r , angle $\theta \rightarrow r(\cos \theta + i \sin \theta)$.